THE TP-NEON: AN INFRA-INSTRUMENTAL CONTRAPTION WHICH MORPHS A MIDI TRUMPET BANDONEON-ERGONOMICS

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ABSTRACT

We describe a hardware-only improvisation and/or compositional set that merges a MIDI trumpet with a circuit bending interface that is inspired on bandoneon ergonomics by using the performer non-dominant hand - normally free in trumped playing. The instrument holistically abides by the principles of infrainstrumental design proposed by Bowens and Archers's [1]. It is termed TP-Neon and its integration within an audio and video analogue modular synthesizer, along with its playability and musicality, is discussed.

1. BACKGROUND

As a reaction to the complexities and interface setup pitfalls of laptop music improvisation, there is a growing number of composers and improvisers interested in increasingly reductionist forms of musical instruments [7]. Busy and dense possibilities of laptop improvisation are driving musicians to consider the integration of modular synthesizers and dedicated hardware equipment into their improvisational and compositional workflows. One step further is the hacking and modifying of existing musical/video devices building simple electronic contraptions. Often, the aim is to research and experiment with surprisingly fresh lo-fi hands-on approaches to sound/video generation [1].

Consequently, there has being a significant influence of lo-fi methods into pop music and academic composition and improvisation. Pop artists such as Bjork, Radio Head and Venetian Snares have readily incorporated these approaches in their recent works [4]. Concomitantly, these lo-fi methods have been featured in performance and composition work by contemporary composers such as, just ti give an example, such as, Nick Collins on his piece The Bowerbird in which several individuals each armed with a loudspeaker and battery make 'Victorian synthetic' sounds [2].

Recently, the concept of lo-fi instrument has being formalized by J. Bowers and P. Archer on his paper, Not hyper, not meta, not cyber but infra-instruments [1]. In this work the authors expand on already established concepts like 'circuit bending' [3,5], or 'hardware hacking [2,6] to define an emerging field of music and video improvisation which they term infrainstruments.

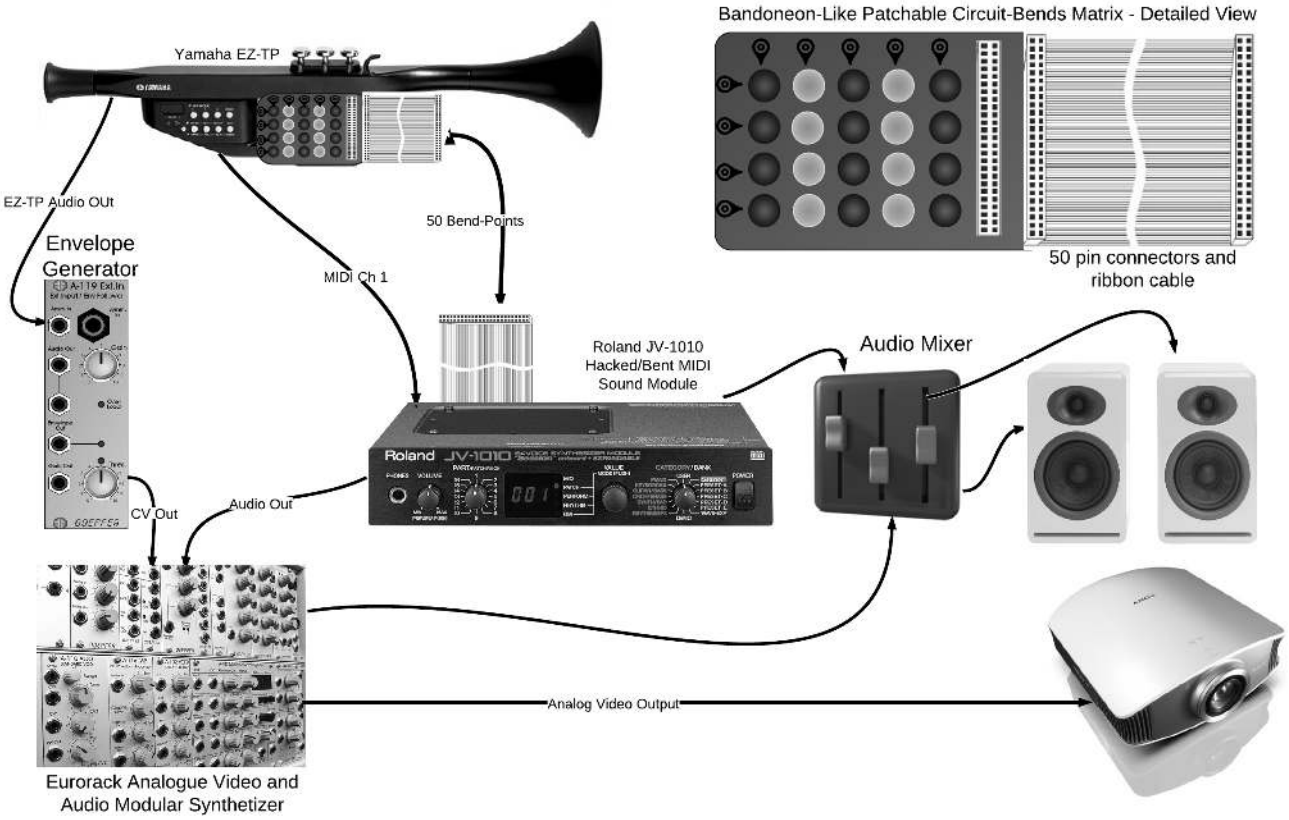
This paper describes a new improvisation set capriciously termed TP-Neon for audio and analogue video that has being inspired by Bowens and Archers's 5-axial definition of infrainstrument, for instance the TP-Neon: (1) is constrained to a limited interactive repertoire; (2) engenders relatively unpredictable music – for instance non-symphonic, non-orchestral; (3) is performed in ways that are predetermined by the architecture of the instrument; (4) is of restricted virtuosity but with a rich set of performance-driven unpredicted explorations and; (5) is ultimately engaging in terms of its visual and sonic appeal, is technically provocative and ergonomically challenging.

2. INSTRUMENT DESCRIPTION

The TP-Neon derives its name from the ergonomic morphing of a MIDI Trumpet (TP) and a Bandoneon-Like DIY control interface (Neon). The figure 1 depicts a numbered TP-Neon architecture. The TP-Neon controller based on a Yamaha EZ-TP MIDI trumpet with a DIY patchable momentary actuation matrix which can engage large number of circuit-bends' combinations with simple presses. The Matrix ergonomically resembles keys on a bandoneon so the TP-Neon instrument can be played with two hands: the non-dominant hand controlling the bend matrix and the trumpet keys played with the dominant hand. From the TP-Neon several signals emerge. The direct audio out from the EZ-TP responds to humming dynamics from the mouthpiece microphone. These are used to generate a control voltage trough an envelope generator which is used ad-hoc within the modular synthetiser. This CV is reused (multiplexed) and modified to generate several daughter CV signals that modulate analogue audio and video signals in the modular. Concomitantly, MIDI messages control a Hacked (Circuit-bent) Roland JV1010 ROM synthetiser module and the audio output is modified in real time by engaging circuit bent combinations. There is a total of 50 bending points that the authors has researched over the past 2 years and all bending points react with each other producing a very large number of bent combinations. This is helped by the DIY bend matrix which consists of momentary switches in a matrix array that can be freely patched to incoming bend points from the JV1010.

3. DISCUSSION AND CONCLUSION

Improvisation with the system results in a rich/complex audiovisual output that is generated from the dynamically controlled CV and audio from the TP-Neon audio processed trough an analogue modular synthesizer (A), EZ-TP MIDI controlled audio from the JV1010 being modulated by the circuit-bending matrix (Neon) combinations. The author has used the TP-Neon live and in his last LP [8] and finds that the technical and musical constrains of the instruments is a powerful creative driving factor. Also, the highly unpredictable character of the instrument when performing enhances the 'happening-effect' and uniqueness of every improvisation. Given that the TP-Neon architecture is reproducible, a notation system can be proposed if more prescriptive compositional works are desired. For instance trumpet notation can be augmented with matrix indexing notation to describe the bend points whilst classic notation of dynamics could be used to describe control voltages at the envelope generator stage. The number of circuit bent combinations from the bandoneon-like keypad allows for combinatorics up to the order of ten to the power of 17 (i.e. 50 bent points with a 20 key patchable switch matrix). In other words, each voice of the JV1010 can be resynthesised to produce around 1.3*1017, a great number considering the simplicity and accessibility of the interface. Also, since humming dynamics at the EZ-TP is converted to continuous control multiplexed voltages in the modular synth, it is possible to control fine nuances of analogue patches with just dynamics alone. For instance, humming softer or louder gives the performer/improviser great control over a number of CV inputs (filter cut-offs, pitches, amplitudes, timing parameters, etc) in real time. This gives the instrument a gratifying sense of nuance control but paradoxically further expends on the unpredictability of results during performance.



4. REFERENCES

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